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|---|---|
| Program : **Diploma in Cloud Computing and Big Data** | |
| Course Code : **6272C** | Course Title: **Big Data Analytics** |
| Semester : **6** | Credits: **4** |
| Course Category: **Open Elective Course** | |
| Periods per week: **4 (L:4 T:0 P:0)** | Periods per semester: **60** |

**Course Objectives:**

- To describe the Big Data concepts
- To explain the basics of data preprocessing and data visualization
- To summarize Big Data tools and techniques.

**Course Prerequisites:**

| Topic | Course Code | Course name | Semester |
|---|---|---|---|
| Understand the concept of Database systems | 3272 | Database System Concepts | 3 |
| Understand the concept of Cloud Computing | 3273 | Introduction to Cloud Computing | 3 |

**Course Outcomes:**

On completion of the course, the student will be able to:

| COn | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| CO1 | Describe the concepts of Big Data | 16 | Understanding |
| CO2 | Summarize the basic concepts in Big Data Analytics and NoSQL | 14 | Understanding |
| CO3 | Demonstrate Map Reduce programming in Hadoop Environment | 16 | Applying |
| CO4 | Use the basic programming tools like Hive and Pig in Hadoop ecosystems | 12 | Applying |
| | Series Test | 2 | |

**CO – PO Mapping**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | | | | | 2 |
| CO2 | 3 | 2 | | | | | 2 |
| CO3 | 2 | 2 | 3 | | | | 2 |
| CO4 | 2 | 2 | 3 | | | | 2 |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline**

| Module Outcomes | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| **CO1** | **Describe the concepts of Big Data** | | |
| M1.01 | Summarize the Characteristics of Big Data | 2 | Understanding |
| M1.02 | Classify Digital Data | 2 | Understanding |
| M1.03 | Describe Big data Analytics and its applications | 2 | Understanding |
| M1.04 | Compare Big Data and Data Warehouse | 1 | Understanding |
| M1.05 | Explain the basics of Data Mining . | 3 | Understanding |
| M1.06 | Illustrate Preprocessing techniques | 3 | Understanding |
| M1.07 | Describe Data Visualization and List some tools used for Visualization | 3 | Understanding |

**Contents:**

**Introduction to Bigdata** : Data –Big data –Examples of Big data

**Characteristics of Big Data**- Volume, Variety, Velocity, and other characteristics.

**Classification /Types of f Digital Data**: Structured, Unstructured and- Semi Structured

**Big Data Analytics :** Definition of Big Data analytics , Requirement of high performance analytics -Challenges with Big Data-Big data analytics used today-Benefits of Big Data analytics

**Applications of Big Data Analytics:** Health care, Education-commerce-Media and entertainment-Finance-Travel industry-Telecom-Automobile

**Data Warehouse**: Definition, Coexistence of Big Data and Data Warehouse.

**Data Mining**- Introduction, Data Mining Goals, Types of data, Steps to discover knowledge from data(Stages of the Data Mining Process), Data Mining Techniques, Knowledge Representation Methods

**Data Preprocessing** -Need of data preprocessing, data quality, data preprocessing methods- data cleaning, data integration, data reduction, data transformation

**Data Visualization:** Definition, Need for Visualization, List some tools used for Visualization

| **CO2** | **Summarize the basic concepts in Big Data Analytics and NoSQL** | | |
|---|---|---|---|
| M2.01 | Explain the importance of Big Data Analytics and the major challenges faced by Big Data . | 3 | Understanding |
| M2.02 | State the definition of Data Science | 1 | Remembering |
| M2.03 | Explain the Responsibilities of a Data Scientist | 1 | Understanding |
| M2.04 | Summarize the Big Data Environment terminologies | 4 | Understanding |
| M2.05 | Describe the concepts of NoSQL | 4 | Understanding |
| M2.06 | Summarize the concepts of NewSQL | 1 | Understanding |
| | Series Test – I | 1 | Understanding |

| | | | |
|---|---|---|---|
| **Contents:**<br>**Big Data Analytics:** Definition, Classification of Analytics, Top Challenges Facing Big Data, Importance of Big Data Analytics, Technologies that meet challenges posed by Big Data.<br>**Data Science:** Definition, Responsibilities of a Data Scientist<br>**Terminologies used in Big Data Environment:** In-memory Analytics, In-Database Processing, Symmetric Multi-Processor Systems, Massively Parallel Processing, Difference Between Parallel and Distributed Systems.<br>**NoSQL:** Definition, Features of NoSQL, Types of NoSQL Databases, Advantages of NoSQL, Drawbacks of NoSQL, Use of NoSQL in Industry,<br>**NewSQL:** Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL | | | |
| **CO3** | **Demonstrate Map Reduce programming in Hadoop Environment** | | |
| M3.01 | Describe the fundamentals of Hadoop Ecosystems. | 4 | Understanding |
| M3.02 | Compare RDBMS and HDFS | 1 | Understanding |
| M3.03 | Explain the components of Hadoop Ecosystem | 3 | Understanding |
| M3.04 | Demonstrate the concept of HDFS and MapReduce | 2 | Applying |
| M3.05 | Use the different function in MapReduce | 6 | Applying |
| **Contents:**<br>**Hadoop-** Features of Hadoop, Key Advantages of Hadoop, Versions, Hadoop Distributions, Hadoop versus SQL, Integrated Hadoop Systems Offered by Leading Market Vendors, Advantages of Hadoop, RDBMS versus Hadoop, Distributed Computing Challenges, Overview of Hadoop Ecosystems and Hadoop Core Components in HDFS.<br>**MapReduce Programming concepts:** Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting, Compression. | | | |
| **CO4** | **Use the basic programming tools like Hive and Pig in Hadoop ecosystems** | | |
| M4.01 | Demonstrate the basic concepts of Hive | 6 | Applying |
| M4.02 | Use the basic concepts of Pig | 6 | Applying |

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|---|---|---|---|
| | Series Test – II | 1 | |

**Contents:**

**INTRODUCTION TO HIVE AND PIG**

**Hive:** Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations. Hive User Defined Function - Serialization and Deserialization (SERDE).

**Pig:** Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview - Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - User-Defined Functions. Pig Versus Hive

**Text / Reference**

| T/R | Book Title/Author |
|---|---|
| T1 | Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 |
| R1 | Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser ‘”Data Mining Concepts and Techniques Third Edition” |
| R2 | Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, Wiley and SAS Business Series, 2012. |
| R3 | Tom White, — Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 |
| R4 | EMC Education Services, Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. |
| R5 | Michael Minelli, Michele Chambers, Ambiga Dhiraj, “Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends”, John Wiley & Sons, 2013. |

**Online Resources**

| Sl.No | Website Link |
|---|---|
| 1 | https://hadoop.apache.org/ |
| 2 | https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ |
| 3 | https://hive.apache.org/ |
| 4 | https://pig.apache.org/ |
| 5 | https://www.tutorialspoint.com/hadoop/index.htm |